

Aliah University

End-Semester Examination (Autumn Semester) - 2022

(For 3rd Year 5th B.Tech (CSE))

Paper Name: Operating Systems

Paper Code: CSEUGPC10

Full Marks: 80

Time: 3 hrs

Group-A (Answer all questions)

1 × 10 = 10

1. What is a time-sharing system?
2. What are operating system services?
3. What is a kernel?
4. What do you mean by inter-process communication?
5. What are privileged instructions?
6. What is meant by context switch?
7. What is the use of fork system call?
8. What is starvation?
9. Define waiting time?
10. Define logical address.

Group-B (Answer any five questions)

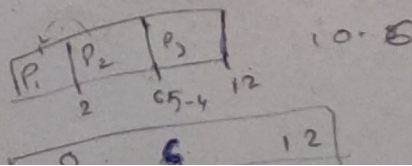
5 × 6 = 30

1. a) Define Operating System. List the objectives of an operating system. b) Explain how multiprogramming increases the utilization of CPU. $4+2=6$
2. a) What is a process? b) With a neat diagram, explain various states of a process. $2+4=6$
3. a) What is synchronization? b) Give Peterson's solution to the Critical section problem. $2+4=6$
4. a) Distinguish between Logical and Physical address space. b) What do you mean by First Fit and best fit memory allocation method? $3+3=6$
5. a) What are the necessary conditions for the occurrence of a deadlock? b) Explain Resource-Allocation graph. $3+3=6$
6. a) What is Belady's Anomaly? b) Define seek time and latency time. c) What are the allocation methods of a disk space? $2+2+2=6$
7. a) Consider a logical address space of 64 pages of 1024 words each, mapped onto a physical memory of 32 frames.
i) How many bits are there in the logical address? ii) How many bits are there in the physical address?
8. Explain about the difference between internal fragmentation and external fragmentation. $3+3=6$

Group-C (Answer any four questions)

4 × 10 = 40

1. a) Explain about various fields of Process Control Block. b) What are the advantages of inter-process communication? c) How communication takes place in a shared-memory environment? Explain. $4+3+3=10$
2. a) What is a Critical Section problem? Give the conditions that a solution to the critical section problem must satisfy. b) What is Dining Philosophers problem? Discuss the solution to the Dining philosopher's problem using monitors. $(2+3)+(3+2)=10$
3. a) What is a page fault? Explain the steps involved in handling a page fault with a neat sketch.
b) Consider the following page reference string: 1,2,3,4,2,1,5,6,2,1,2,3,7,6,3,2,1,2,3,6
How many page faults would occur for the Optimal and LRU page replacement algorithm, assuming three frames and all frames are initially empty. $2+8=10$



4. a) What is a deadlock? How are deadlocks detected?

b) Consider the following snapshot of a system:

Process	Allocation	Max	Available
	A B C D	A B C D	A B C D
P0	0 0 1 2	0 0 1 2	1 5 2 0
P1	1 0 0 0	1 7 5 0	
P2	1 3 5 4	2 3 5 6	
P3	0 6 3 2	0 6 5 2	
P4	0 0 1 4	0 6 5 6	

Answer the following questions using banker's algorithm:

- What is the content of the matrix Need?
- Is the system in a safe state?

$$(2+3) + 5 = 10$$

5. a) Explain and compare the SCAN and C-SCAN disk scheduling algorithms.

b) Suppose that a disk drive has 5000 cylinders, numbered 0 to 4999. The drive is currently serving a request at cylinder 143, and the previous request was at cylinder 125. The queue of pending requests, in FIFO order is:

86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130

Starting from the current head position, what is the total distance that the disk arm moves to satisfy all the pending requests for FCFS and SSTF disk-scheduling algorithms?

$$3+7=10$$

6. a) Define CPU Scheduling. b) Distinguish between preemptive and non-preemptive Scheduling. c) What are the various scheduling criteria for CPU scheduling?

d) Consider the following table and find out i) average waiting time ii) turnaround time for preemptive-SJF CPU scheduling algorithms.

Process	Arrival Time	Burst Time
P1	0	7
P2	1	3
P3	2	4
P4	3	2

$$2+3+2+3=10$$

$$\begin{array}{r} 013 \\ 143 \\ \hline 86 \\ 37 \end{array}$$

$$\begin{array}{r} 1620 \\ 728 \\ 487 \\ 561 \\ 826 \\ 861 \\ 557 \\ 1287 \\ 57 \\ \hline 7081 \end{array}$$

$$\begin{array}{r} 15000 \\ - 12 \\ \hline 14988 \end{array}$$